

Automatidata | Project Proposal

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How I approached the problem

I treated the dataset as three connected analytical problems rather than forcing everything into one model.

First, I tested the observed fare difference between payment groups using Welch's t-test and a positive-fare sensitivity analysis.

Second, I built a fare-estimation workflow comparing linear and tree-based models. Model selection was performed on validation data, while the final test set remained untouched until the end.

Third, I built a generous-tipping classifier and evaluated it with precision, recall, F1, ROC AUC, and a confusion matrix.

What the analysis produced

- Statistical evidence of a difference in observed mean fares between payment groups, with an estimated difference of about \$1.22.
- A selected Gradient Boosting fare model with test MAE of \$3.55, RMSE of \$16.41, and R^2 of 0.182.
- A selected Random Forest tipping classifier with test recall of 0.731 and F1 of 0.562.
- Diagnostic analysis showing where fare-prediction errors concentrate.
- Clear documentation of data limitations, leakage considerations, and production-readiness requirements.

Why this workflow matters

The goal was not simply to produce a high model score. The workflow keeps the statistical question, model-selection process, final evaluation, and business interpretation separate so the results can be understood and challenged.

This project is a portfolio case study based on an educational dataset. It does not claim prior client work for Automatidata.